

Fig. 4: A single-side, actual-size PCB layout for PIC16F877A-based temperature monitoring system

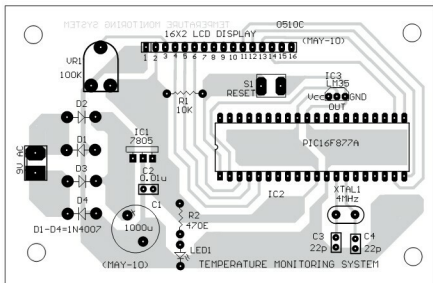


Fig. 5: Component layout for the PCB

extension. You can create source files `tempr.c`, `delay.c` and `lcd.c` in this manner. After typing and saving the code, you have to add the source files to your project. In 'Project' menu, click 'Add Files to Project' and then add the files by browsing them from the location they are saved in.

3. *Configure the system.* To configure features like type of oscillator and WDT, click 'Configure' menu and then click 'Configuration Bits.' In the configuration bits window, select

the type of oscillator as XT, and disable all other features like watchdog timer, power-up timer and brownout detect.

4. *Compile the project.* To compile the software, click 'Build' option in 'Project' menu. The software is compiled and 'Build Successful' message appears in the output window. After successful compilation of the program, the file *tempr.hex* is generated.

Downloading the software into the chip. After compiling the software, the hex file generated can be

downloaded into the chip either using a locally made programmer (like JMD programmer) or Microchip's programming kit (PICKit2, PICKit3, ICD2, ICD3, etc).

We have used Microchip MPLAB ICD2 (in-circuit debugger) to burn the program into PIC16F877A microcontroller.

The programming can be done from MPLAB IDE itself by selecting the kit (MPLAB ICD2) from 'Programmer' menu in the menu bar. If you use a JMD programmer, the hex file can be downloaded into the chip using WinPic800 software. The configuration bits have to be separately configured in WinPic800.

Construction

A single-side, solder-side PCB layout of the temperature monitoring system is shown in Fig. 4 and its component layout in Fig. 5.

A 230V AC primary to 0-9V AC secondary step-down transformer is used to step down the 230V mains supply. The output of the transformer is rectified using a bridge rectifier consisting of diodes D1 through D4 and regulated to 5V using 7805 voltage regulator. The glowing of the LED (LED1) indicates the presence of 5V supply in the circuit.

The analogue output of the temperature sensor LM35 (IC3) from its pin 2 is connected to RA0/AN0 pin of PIC16F877A microcontroller (IC2). A 4MHz crystal (X_{TAL1}) is connected to pins 13 and 14 of the microcontroller. The 16x2 LCD is interfaced with the microcontroller using Port A and Port D. The control signals for the LCD are provided using Port-A pins RA1, RA2 and RA3. The command and data for the LCD are provided using Port-D pins RD0 through RD3. The remaining power supply connections are as shown in Fig. 2.

EFY note. The source code of this article has been included in this month's EFY-CD. It is also available on www.efymag.com website. ●